

Manufacturing Operational Insights Report

on

Marketing Content Generator

Course Name: **Generative AI**

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2025-2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used *(If applicable)*

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

1. Problem Statement

Marketing professionals often struggle to generate high-quality, platform-specific content quickly while maintaining brand consistency across different channels. There is a need for an automated system that can transform marketing topics into professional copy and visual assets while "remembering" previous brand history to ensure a cohesive voice.

2. Project Objectives

- Develop a GenAI-powered web application to transform marketing topics into professional content.
- Implement a "Brand Memory" system using Retrieval Augmented Generation (RAG) to maintain tone and terminology consistency.
- Integrate multi-modal capabilities, including high-speed text generation and photorealistic image generation.
- Provide a user-friendly interface for generating ad copy, social media posts, emails, and product descriptions.

3. Scope of the Project

The project covers the high-level design and deployment of the Marketing Content Generator, including:

- **System Architecture:** A 5-layer model spanning User Zone to External Services .
- **Content Generation:** Automated text creation using AIDA, PAS, FAB, and Hook-Story-CTA frameworks.
- **Image Generation:** Integration with FLUX.1-schnell for professional 4K marketing visuals.
- **Data Management:** Vector database implementation using ChromaDB for semantic search and brand context.

Proposed Solution

1.Key Features

- **Multi-Platform Content:** Specialized templates for Ad Copy, Social Media, Emails, and Product Descriptions.
- **Brand Memory (RAG):** Injects semantically similar past content into new prompts to ensure brand consistency.
- **AI Image Generator:** Creates 1024x1024 photorealistic images based on the generated marketing text.
- **Export Functionality:** Capability to download generated content in .txt or .docx formats.
- **Live History:** A sidebar library to browse and filter recent generations.

2.Overall Architecture / Workflow

The application follows a **layered monolithic architecture**:

1. **Presentation Layer:** Streamlit-based web interface for user inputs and output rendering.
2. **Business Logic Layer:** LLM and Image engines that process data and prompt templates.
3. **Data Access Layer:** Manages communication between the logic layer and the vector store.
4. **Persistence Layer:** ChromaDB (4 collections) for storing embeddings and metadata.
5. **External Services:** API integrations with Groq (LLM) and Hugging Face (Image Generation).

USER ZONE

Web Browser

Chrome / Edge / Firefox

User Inputs

6 fields: Topic, Audience, Tone, Platform, Type, USP

↓ HTTP to localhost:8501 (local) or HTTPS (Streamlit Cloud)

PRESENTATION LAYER

ui/app.py — Streamlit Web Interface

Page config, CSS theme, sidebar, form inputs, output rendering, session state

↓ Python function calls

BUSINESS LOGIC LAYER

LLM Engine

core/llm_engine.py

Image Engine

core/image_engine.py

Prompt Templates

prompts/templates.py

↓ data operations

DATA ACCESS LAYER

Vector Store

vectordb/store.py

File Exporter

utils/exporter.py

↓ read/write

PERSISTENCE LAYER

ChromaDB — 4 Collections

ad_copy | social_media | email_campaign | product_description

↓ HTTPS outbound API calls

EXTERNAL SERVICES

Groq API

llama-3.3-70b-versatile 500 tokens/sec

Hugging Face API

FLUX.1-schnell Photorealistic images

Fig.1: System Architecture Diagram — 5 Layer Model

Process Flow

The system supports two primary process flows. Both are triggered by user interaction in the Streamlit interface.

Process Flow 1 — Text Content Generation

START: User fills 6 form fields and clicks Generate Content

↓

Input Validation

Check: topic, audience, USP non-empty. If invalid: st.error() shown, process stops.

↓ Inputs valid

ChromaDB Search

`find_similar(content_type, topic, audience)`

Similarity > 30%?

YES: inject brand context NO: empty context string

↓ similar_context string

Prompt Engineering

Select AIDA/PAS/FAB template. Inject: topic, audience, tone, platform, USP, brand context.

↓ System + User message pair

Groq LLM API Call

`llama-3.3-70b-versatile | max_tokens=1024 | temperature=0.75`

↓ Generated text + token count

Save to ChromaDB

Embed + store with metadata

Update Session State

generated_content, generation_meta

↓ st.rerun()

END: Content card displayed with badges, export buttons, image section

Process Flow 2 — AI Image Generation

START: User clicks Image Prompt button

↓

Groq: generate_image_prompt()

Single user-turn | max_tokens=300 | temp=0.85 | IMAGE_PROMPT_TEMPLATE

↓ Image prompt stored in session state. Generate Image button enabled.

User clicks Generate Image

↓

build_marketing_prompt()

Combines image_prompt + quality suffix: photorealistic, 4K, professional photography

↓

Hugging Face POST

FLUX.1-schnell | 4 inference steps | guidance=0.0 | 1024x1024 | timeout=120s

↓ Raw PNG bytes

END: Image rendered + Download Image button shown

Fig.2: Process Flow

Information Flow

This table shows how data transforms as it moves through the system from raw user inputs to final outputs.

Stage	Input	Output
User Input	Raw form fields (strings)	Validated Python variables: 6 typed strings
Vector Search	topic + audience strings	similar_context: formatted past brand content or empty string
Prompt Building	6 inputs + similar_context	Structured prompt: system + user messages (500-800 tokens total)
LLM Generation	Structured prompt	Generated content string (300-800 tokens) + integer token count
Vector Storage	Content + metadata	ChromaDB doc: UUID id + vector[384] + metadata dict
Image Prompt Gen	content_type, topic, tone, platform	Detailed image prompt string (150-300 words)
Image Generation	Image prompt + quality suffix	Raw PNG bytes (1024x1024, approx 2-4MB)

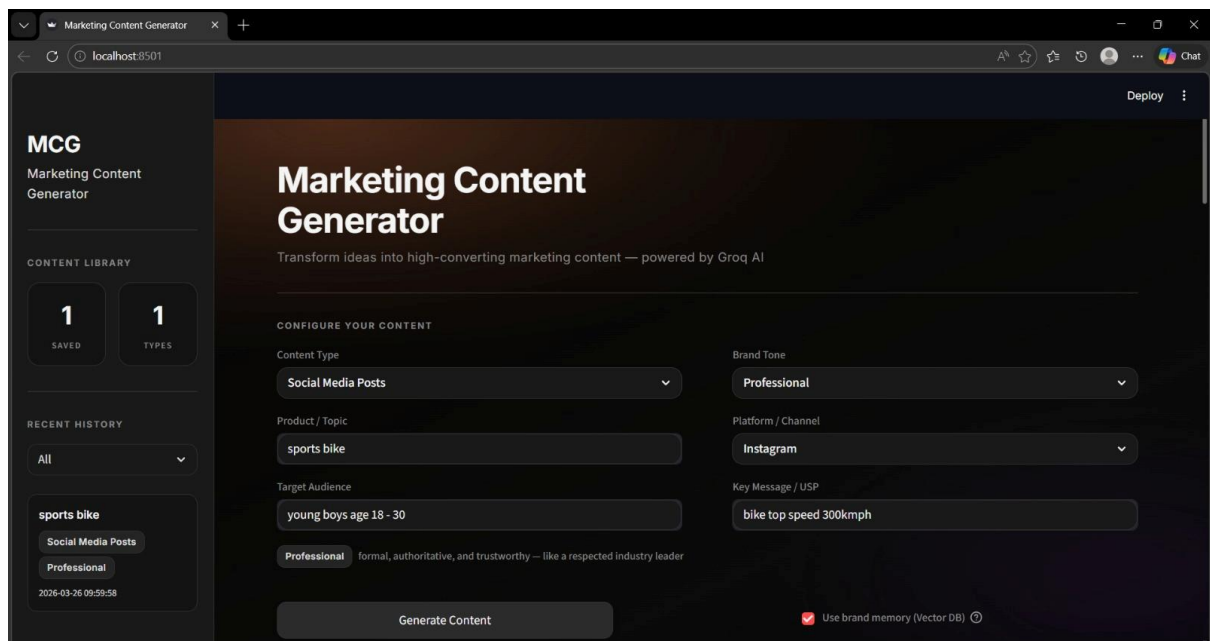
Stage	Input	Output
Export	Generated content + metadata	Plain text string (.txt) or binary bytes (.docx)

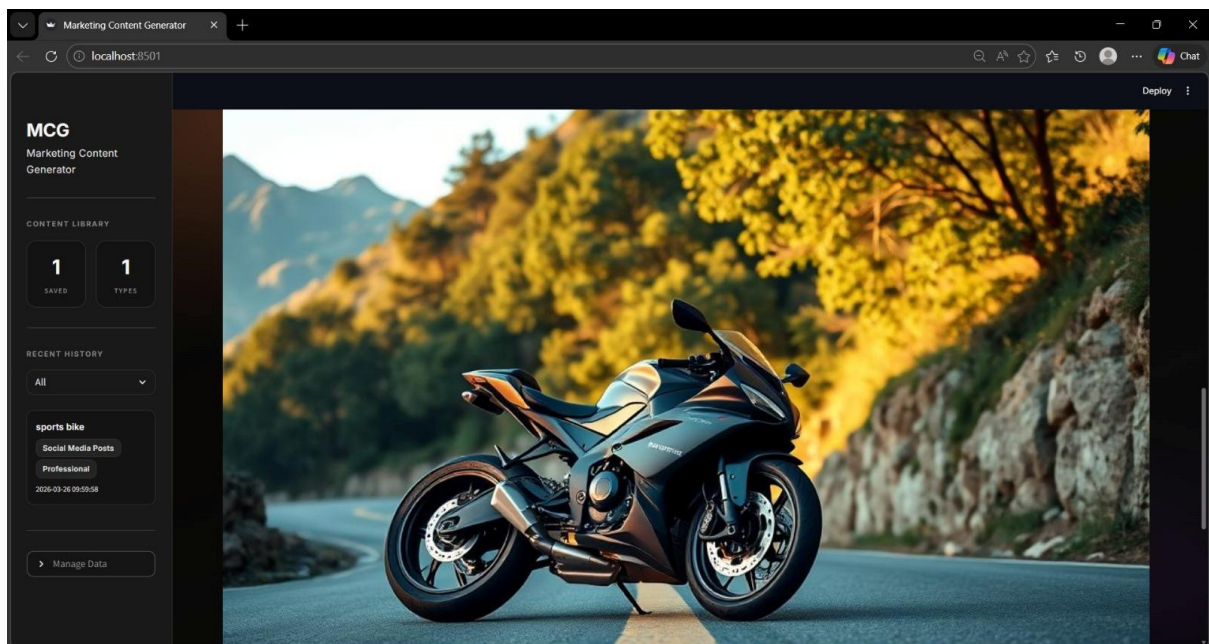
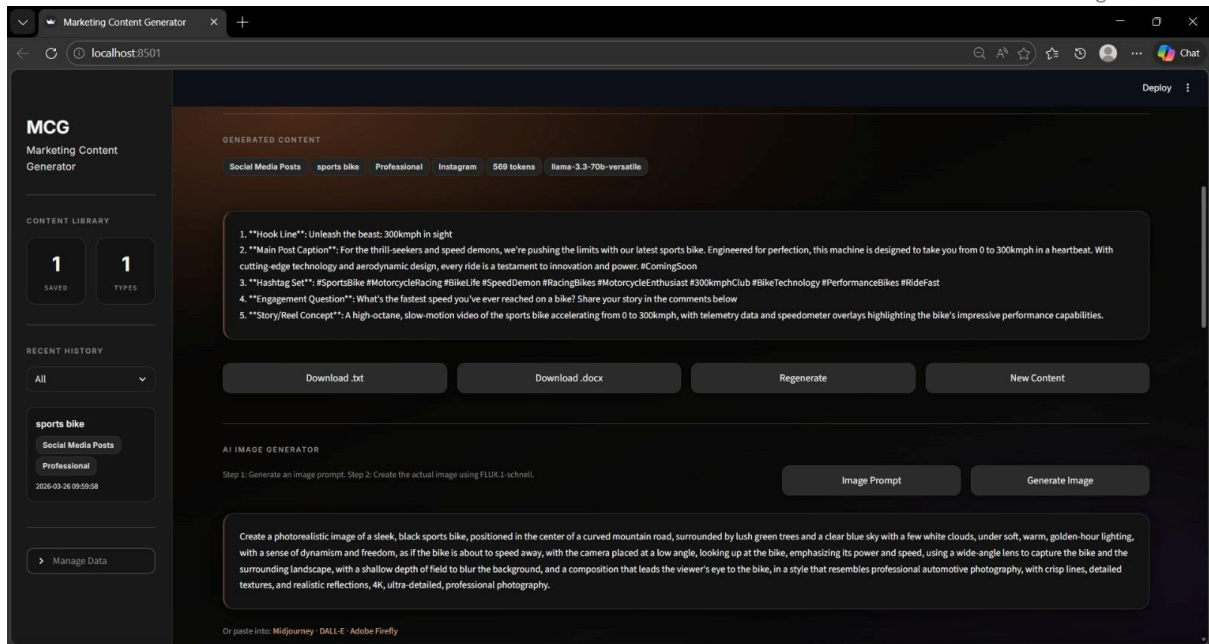
3.Tools & Technologies Used

- **Language:** Python
- **Large Language Model:** Groq (llama-3.3-70b-versatile)
- **Vector Database:** ChromaDB
- **Image Model:** FLUX.1-schnell (via Hugging Face API)
- **Web Framework:** Streamlit
- **Deployment:** Streamlit Cloud and GitHub

Results & Output

1. Screenshots / Outputs





2. Reports / Dashboards / Models

The Marketing Content Generator utilizes three specialized AI models to handle text, images, and data retrieval:

- **Large Language Model (LLM):** Uses llama-3.3-70b-versatile via the Groq API for high-speed text generation (500 tokens/sec) and crafting image prompts.
- **Image Generation Model:** Uses FLUX.1-schnell via Hugging Face to produce 1024x1024 photorealistic marketing visuals in 15–40 seconds.
- **Embedding Model:** Uses all-MiniLM-L6-v2 to convert content into 384-dimension vectors for semantic search and brand memory within ChromaDB.

Would you like me to explain how the system uses these models to maintain brand consistency

3. Key Outcomes

The following are the key outcomes of the Marketing Content Generator project:

- **High-Speed Generation:** Achieved text generation at 500 tokens/sec using the Groq llama-3.3-70b-versatile model.
- **Brand Consistency:** Successfully implemented Brand Memory via RAG, using semantic search to maintain tone across all generations .
- **Multi-Modal Output:** Automated the creation of both professional copy and 1024x1024 photorealistic images.
- **Efficient Architecture:** Developed a 5-layer modular system that separates business logic from data access and external APIs .
- **Production Readiness:** Fully deployed on Streamlit Cloud with secure API key management and automatic redeployment.

Conclusion

The **Marketing Content Generator** successfully demonstrates a high-performance, GenAI-powered solution for modern marketing needs. By integrating **Groq LLMs** for rapid text generation and **FLUX.1-schnell** for professional visuals, the system provides a comprehensive multi-modal experience.

The project's key achievement is the implementation of **Brand Memory via RAG**, which uses **ChromaDB** to ensure that all generated content remains consistent with a brand's established tone and history. This modular architecture allows for easy scalability, while the deployment on **Streamlit Cloud** ensures a secure, production-ready environment. Ultimately, the project fulfills its goal of automating complex creative workflows while maintaining high standards of performance and architectural integrity.

Future Scope & Enhancements

- **Persistent Cloud Storage:** Transition from the current ephemeral local ChromaDB storage to **Pinecone**, a hosted vector database, to ensure user data persists across cloud restarts.
- **Advanced Data Migration:** Implement a one-time migration script to export local collections to JSON and import them into **Pinecone indexes** while maintaining the existing document schema.
- **Multi-User Support:** Evolve the system from a single-user local deployment to a multi-user platform with coordinated database migrations and session handling.
- **Enhanced Model Integration:** Continue monitoring for **Groq and Hugging Face model deprecations** to ensure the system always utilizes the most current and efficient LLM and image generation models.